

EASTERN AFRICA STATISTICAL TRAINING CENTRE



TITLE OF THE PROJECT

Predicting Under-Five Child Mortality in Tanzania Using Demographic and Household Characteristics

PROGRAMME: BACHELOR DEGREE IN DATA SCIENCE - 2025/2026

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CHAPTER ONE: INTRODUCTION

1. Background of the study

Under-five child mortality remains a critical public health and development challenge globally, with the highest burden occurring in low- and middle-income countries.

According to the World Health Organization, millions of children die each year before reaching their fifth birthday, largely due to preventable causes such as infectious diseases, malnutrition, and inadequate access to healthcare services (World Health Organization, 2021). Sub-Saharan Africa accounts for a substantial proportion of these deaths, reflecting persistent inequalities in child health outcomes across regions.

In Tanzania, considerable progress has been made in reducing under-five mortality over the past two decades. However, recent evidence shows that the mortality rate remains a public health concern, particularly in rural areas and among socioeconomically disadvantaged populations (National Bureau of Statistics (NBS) & ICF, 2023). Factors such as low maternal education, poor household living conditions, limited access to safe water and sanitation, and inadequate healthcare services continue to contribute to preventable child deaths. These patterns suggest that improvements in national averages may mask significant disparities across regions and population groups.

Previous empirical studies have identified several determinants of under-five mortality, including household wealth, maternal characteristics, environmental conditions, and access to healthcare services (World Health Organization, 2021). Most of these studies have relied on traditional statistical approaches such as logistic regression, which are effective in identifying associations between variables but may have limitations in capturing complex and nonlinear relationships among multiple risk factors. As a result, their ability to accurately predict mortality risk at the individual level remains limited.

The increasing availability of large-scale, nationally representative datasets, such as the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey (TDHS-

MIS), provides an opportunity to apply advanced data science techniques to child mortality analysis (National Bureau of Statistics (NBS) & ICF, 2023). Machine learning methods, in particular, offer the ability to model complex interactions, improve predictive accuracy, and identify high-risk populations more effectively than conventional approaches (Breiman, 2001).

Therefore, applying predictive analytics to under-five mortality in Tanzania can provide deeper insights into risk patterns and strengthen evidence-based decision-making. This approach has the potential to support policymakers and health planners in designing targeted interventions aimed at reducing preventable child deaths and improving child health outcomes in line with national and global development goals.

1.2 Statement of the Problem

Despite notable progress in reducing under-five mortality in Tanzania, child deaths remain relatively high, particularly among vulnerable and socioeconomically disadvantaged populations. According to the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey, disparities in child survival persist across regions, households, and population groups (National Bureau of Statistics (NBS) & ICF, 2023). These disparities indicate that existing interventions have not fully addressed the underlying risk factors affecting child survival.

Previous studies conducted in Tanzania and similar contexts have primarily focused on identifying associations between socioeconomic, demographic, and environmental factors and under-five mortality using traditional statistical methods. While these approaches provide valuable insights into determinants of child mortality, they are largely limited to explanatory analysis and do not adequately support accurate prediction of high-risk cases at the individual or household level.

The lack of robust predictive models limits the ability of policymakers and health planners to efficiently target interventions and allocate resources to populations most at risk. Without precise identification of high-risk children, interventions may remain generalized rather than targeted, thereby reducing their effectiveness in addressing preventable child deaths.

Furthermore, although nationally representative datasets such as TDHS-MIS 2022 are available, their potential has not been fully utilized through advanced data science techniques. In particular, the application of machine learning methods in predicting under-five mortality in Tanzania remains limited, and there is insufficient comparative evidence regarding their performance relative to traditional statistical models (Breiman, 2001).

Therefore, there is a need to develop predictive models that utilize demographic, maternal, and household characteristics to accurately identify children at high risk of mortality. Such models can enhance data-driven decision-making and support the design of targeted, efficient, and evidence-based interventions aimed at reducing under-five child mortality in Tanzania.

1.3 General Objective

To predict under-five child mortality in Tanzania.

1.4 Specific Objectives

1. To identify demographic, maternal, and household factors associated with under-five child mortality in Tanzania.
2. To develop a logistic regression model for predicting under-five child mortality.
3. To compare the predictive performance of logistic regression and machine learning models.

1.5 Research Questions

1. Which demographic, maternal, and household factors are associated with under-five child mortality in Tanzania?
2. How well does a logistic regression model predict under-five child mortality?
3. How does the predictive performance of logistic regression compare to machine learning models (Decision Tree, Random Forest, and Gradient Boosting) based on accuracy, precision, recall, F1-score, and AUC-ROC?

1.6 Significance of the Study

This study is expected to contribute both academically and practically to the field of public health and data science. From an academic perspective, it demonstrates the application of advanced data science techniques, including statistical modeling and machine learning, in analyzing and predicting under-five child mortality using nationally representative data. The study also contributes to existing literature by providing comparative evidence on the performance of traditional statistical models and machine learning approaches within the Tanzanian context.

From a practical perspective, the findings of this study will support policymakers, health planners, and development partners in identifying high-risk populations based on data-driven insights. By enabling more accurate prediction of under-five mortality risk, the study can inform the design and implementation of targeted and efficient interventions. This has the potential to improve resource allocation, enhance the effectiveness of child health programs, and contribute to reducing preventable child deaths in Tanzania.

Furthermore, the study aligns with national health priorities and global development frameworks, particularly Sustainable Development Goal 3, which aims to ensure healthy lives and promote well-being for all. By integrating predictive analytics into public health decision-making, the study provides a foundation for more evidence-based and technologically informed approaches to improving child survival outcomes.

1.7 Scope of the Study

This study focuses on under-five children in Tanzania using cross-sectional data from the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey (TDHS-MIS). The data covers all regions of mainland Tanzania and Zanzibar.

The analysis is restricted to variables available in the Births Recode (BR) dataset, including:

- Demographic factors (child's sex, birth order)
- Maternal factors (age, education, antenatal care)
- Household socioeconomic conditions (wealth index, water source, sanitation)
- Geographic information (region, place of residence)

The study is limited to secondary data analysis and does not involve primary data collection. As a cross-sectional study, it does not establish causal relationships but focuses on identifying patterns and predicting the risk of under-five child mortality. Findings are intended for predictive and risk profiling purposes, not causal inference.

The study is confined to the following modeling techniques: logistic regression, Decision Tree, Random Forest, and Gradient Boosting. Results may be influenced by data quality, missing values, and limitations inherent in survey-based data (e.g., recall bias, social desirability bias).

CHAPTER TWO: LITERATURE REVIEW

2.1.1 Under-Five Mortality

Under-five mortality refers to the death of a child before reaching five years of age. It is commonly measured as the probability of dying before age five per 1,000 live births (World Health Organization, 2021). In this study, under five mortality is operationalized as a binary outcome variable derived from child survival status in the Tanzania Demographic and Health Survey (TDHS-MIS), where a value of 1 indicates death before five years of age and 0 indicates survival.

2.1.2 Demographic Factors

Demographic factors refer to characteristics that describe the structure and composition of a population, including variables such as child's age, sex, birth order, and place of residence (urban or rural). These factors are important determinants of child health outcomes and are commonly used in analyzing variations in under-five mortality (National Bureau of Statistics (NBS) & ICF, 2023).

2.1.3 Household Characteristics

Household characteristics refer to the socioeconomic and environmental conditions within a household that may influence child health and survival. These include variables such as household wealth index, source of drinking water, sanitation facilities, and housing conditions. Such factors are associated with exposure to health risks and access to basic services (World Health Organization, 2021).

2.1.4 Predictive Analytics

Predictive analytics refers to the use of statistical and machine learning techniques to analyze historical data and generate predictions about future or unknown outcomes (James et al., 2021). In this study, predictive analytics is applied to model and estimate the risk of under-five child mortality using demographic, maternal, and household variables, with the aim of identifying high-risk populations.

2.2 Theoretical Framework

This study is guided by the Social Determinants of Health (SDH) Theory, which emphasizes that health outcomes are largely influenced by social, economic, and environmental conditions rather than solely by biological factors (World Health Organization, 2021). The theory posits that differences in income, education, living conditions, and access to essential services create inequalities in health outcomes across populations.

Within the context of under-five child mortality, the SDH framework suggests that a child's survival is shaped by multiple interacting factors at the individual, household, and community levels. For example, maternal education influences health knowledge and care-seeking behavior, household wealth affects access to nutrition and healthcare services, and environmental conditions such as water and sanitation determine exposure to infectious diseases. These interrelated factors collectively determine the risk of child mortality (National Bureau of Statistics (NBS) & ICF, 2023).

The application of this theory provides a conceptual basis for selecting key independent variables in this study, including demographic characteristics (e.g., child's sex and birth order), maternal factors (e.g., maternal age and education), household socioeconomic conditions (e.g., wealth index, water source, and sanitation), and geographic factors (e.g., place of residence and region).

Furthermore, the SDH framework supports the use of predictive modeling approaches by recognizing that health outcomes are influenced by complex and potentially nonlinear interactions among multiple determinants. Machine learning methods are therefore appropriate, as they can capture these complex relationships more effectively than traditional statistical models (Breiman, 2001).

By grounding the study in the Social Determinants of Health Theory, the research ensures that both variable selection and analytical methods are theoretically justified and aligned with established public health principles.

2.3 Empirical Review

Empirical studies across sub-Saharan Africa have consistently identified a range of socioeconomic, demographic, and environmental factors as key determinants of under-five child mortality. Several studies using Demographic and Health Survey (DHS) data have applied logistic regression models to examine these relationships. Findings commonly indicate that children born to mothers with low levels of education and those living in poorer households face significantly higher risks of mortality. Maternal education, in particular, has been shown to influence health-seeking behavior, child care practices, and access to healthcare services, thereby playing a critical role in child survival outcomes (World Health Organization, 2021).

In addition to socioeconomic factors, environmental conditions such as access to clean water and improved sanitation have been widely documented as significant contributors to child health. Studies conducted in multiple African countries report that inadequate water and sanitation facilities increase exposure to infectious diseases such as diarrhea, which is a leading cause of under-five mortality. For instance, Worede et al. (2025) found that poor sanitation and unsafe drinking water are strongly associated with higher prevalence of childhood illnesses and increased mortality risk.

While these studies provide important insights into the determinants of under-five mortality, they are largely based on traditional statistical methods, particularly logistic regression. These methods are effective in identifying associations between variables but are limited in their ability to capture complex, nonlinear relationships and interactions among multiple risk factors. As a result, their predictive performance at the individual level may be constrained.

Recent studies in the field of public health have begun to explore the application of machine learning techniques, such as decision trees, random forests, and gradient boosting, to improve predictive accuracy. These methods offer advantages in handling high-dimensional data, modeling nonlinear relationships, and automatically capturing interactions between variables (Breiman, 2001; James et al., 2021). However, their application to under-five mortality analysis using recent Tanzanian data remains limited.

Furthermore, there is a lack of comparative studies evaluating the performance of traditional statistical models against machine learning approaches within the same analytical framework. This creates a gap in understanding whether advanced predictive methods provide meaningful improvements in identifying high-risk populations.

Therefore, this study seeks to address these gaps by applying both logistic regression and machine learning models to nationally representative TDHS-MIS 2022 data, and by systematically comparing their predictive performance. This approach contributes to existing literature by integrating traditional and modern analytical techniques to enhance the prediction of under-five child mortality in Tanzania.

1.2 Research Gap

Although numerous studies have examined the determinants of under-five mortality in sub-Saharan Africa, including Tanzania, most existing research relies primarily on traditional statistical methods such as logistic regression and focuses on explanatory relationships. While these approaches identify significant associations, they often have limited predictive capacity and may not fully capture complex nonlinear interactions among variables.

Furthermore, few studies in Tanzania have applied machine learning techniques to recent Demographic and Health Survey (DHS) data to predict under-five mortality risk. Comparative evaluations between conventional statistical models and machine learning models remain limited. As a result, there is insufficient evidence on whether advanced predictive methods can improve identification of high-risk children and support targeted public health interventions.

This study seeks to address this gap by developing and comparing statistical and machine learning models using nationally representative 2022 TDHS-MIS data.

1.3 Conceptual Framework

Under-five child mortality is considered the dependent variable in this study. It is influenced by multiple categories of independent variables derived from the Social Determinants of Health framework. These include demographic factors (such as child's age, sex, and birth order), maternal characteristics (including maternal education, age, and healthcare utilization), household socioeconomic conditions (such as wealth index, access to clean water, sanitation facilities, and housing conditions), and geographic location (urban–rural residence and region).

These factors are hypothesized to directly or indirectly affect child survival outcomes. The conceptual framework assumes that socioeconomic and environmental conditions shape exposure to health risks, which in turn influence mortality probability. This structure guides variable selection and model development for predicting under-five mortality risk.

CHAPTER THREE: RESEARCH METHODOLOGY

3.0 Study area

This study is conducted in the United Republic of Tanzania, located in East Africa. Tanzania has a diverse population distributed across urban and rural areas, with significant regional variations in socioeconomic and health conditions. According to national statistics, Tanzania continues to experience challenges related to child health, including preventable under-five mortality.

The analysis utilizes nationally representative data from the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey (TDHS-MIS), which covers all regions of mainland Tanzania and Zanzibar. The survey collects comprehensive information on demographic characteristics, maternal health, household conditions, and child survival outcomes, making it suitable for examining predictors of under-five mortality.

3.1 Study design

This study employs a quantitative, cross-sectional research design using secondary data from the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey (TDHS-MIS). The TDHS-MIS is a nationally representative household survey conducted using a stratified two-stage cluster sampling technique.

The cross-sectional design allows for the analysis of relationships between under-five child mortality and selected demographic, maternal, household, and geographic variables at a single point in time. Since the study utilizes existing survey data, it does not involve primary data collection.

3.2 Population of the study

This study utilizes secondary data obtained from the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey (TDHS-MIS). The TDHS-MIS is a nationally representative household survey conducted by the National Bureau of Statistics (NBS) in collaboration with the Ministry of Health and international development partners. The

survey provides comprehensive information on demographic characteristics, maternal and child health, fertility, mortality, and household socioeconomic conditions.

The TDHS-MIS employed a stratified two-stage cluster sampling design to ensure national and regional representativeness. In the first stage, enumeration areas (EAs) were selected from the national sampling frame using probability proportional to size. In the second stage, households were systematically selected within each sampled enumeration area. Stratification was performed by region and urban rural residence to enhance the precision and representativeness of the estimates.

Sampling weights were provided in the dataset to account for unequal probabilities of selection and non-response. These weights will be applied in the analysis to generate unbiased and nationally representative estimates.

For this study, the Births Recode (BR) dataset (TZBR81FL) is used. The BR file contains detailed birth history and child survival information for children born to interviewed women, along with linked maternal and household characteristics, making it appropriate for modeling under-five child mortality.

3.3 Sample size and sampling techniques

The study population consists of children born to interviewed women included in the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey. The analysis is based on the Births Recode (BR) dataset, which contains information on all recorded births within the survey reference period.

For this study, children under five years of age at the time of the survey will be included in the analysis. Records with incomplete information on key variables required for modeling will be excluded. The final analytical sample size will be determined after data cleaning and application of inclusion criteria.

3.4 Variables and Measurement

(a) Dependent Variable

The dependent variable, under-five mortality, is derived from the birth history variables indicating child survival status (b5) and age at death (b7). A binary outcome variable will be created, where 1 represents death before reaching five years of age (less than 60 months), and 0 represents survival beyond five years.

(b) Independent Variables

Independent variables are selected based on the Social Determinants of Health framework and prior empirical literature. These variables are categorized into four groups:

- 1) Demographic variables: child's sex, birth order, and place of residence (urban/rural).
- 2) Maternal characteristics: maternal age at birth, maternal education level, and antenatal care utilization.
- 3) Household socioeconomic variables: household wealth index, source of drinking water, and type of sanitation facility.
- 4) Geographic variables: region and area of residence.

Categorical variables will be coded using appropriate reference categories for regression modeling. Continuous variables, where applicable, will be retained in their original scale or categorized based on meaningful thresholds.

3.5 Data Analysis Methods

Data analysis will be conducted using Python statistical and machine learning libraries. The analysis will follow a systematic and reproducible workflow, beginning with data preprocessing and exploratory analysis, followed by statistical modeling, machine learning implementation, and comparative model evaluation. The analytical process is designed to achieve two main objectives: (i) to identify significant predictors associated with under-five child mortality, and (ii) to assess the predictive performance of statistical and machine learning models. All procedures will be documented to ensure transparency and methodological rigor.

The analysis will integrate both inferential statistical techniques and predictive modeling approaches. Logistic regression will be used as a baseline statistical model to estimate associations between predictors and mortality risk. Subsequently, machine learning algorithms will be applied to evaluate whether advanced modeling techniques improve prediction accuracy and risk classification.

Sampling weights provided in the TDHS-MIS dataset will be incorporated where appropriate to maintain national representativeness. Model assumptions, performance metrics, and validation procedures will be carefully examined to ensure reliability and robustness of results.

3.6.1 Data Preparation

The dataset will be cleaned to remove incomplete or inconsistent records. Missing values in key variables will be handled using appropriate techniques such as exclusion or, where justified, imputation. Sampling weights provided in the TDHS-MIS dataset will be applied to account for unequal probabilities of selection and non-response, thereby ensuring nationally representative estimates.

Categorical variables will be encoded appropriately for modeling purposes. The dependent variable will be constructed as a binary outcome variable representing under-five mortality status.

3.6.2 Descriptive Analysis

Descriptive statistics will be computed to summarize the demographic, maternal, household, and geographic characteristics of the study population. Frequencies and percentages will be reported for categorical variables, while means and standard deviations will be presented for continuous variables. This stage will provide an overview of the distribution of predictors and mortality outcomes.

3.6.3 Statistical Modeling

A baseline logistic regression model will be developed to examine associations between selected independent variables and under-five mortality. Logistic regression is appropriate due to the binary nature of the dependent variable. Adjusted odds ratios and confidence intervals will be estimated to assess the strength and direction of relationships.

3.6.4 Machine Learning Modeling

To enhance predictive performance, selected machine learning algorithms including Decision Tree, Random Forest, and Gradient Boosting classifiers will be implemented. These models are capable of capturing nonlinear relationships and complex interactions among predictors that may not be fully addressed by traditional regression methods.

The dataset will be divided into training (80%) and testing (20%) subsets to evaluate out-of-sample predictive performance. To enhance model stability and reduce the risk of overfitting, 5-fold cross-validation will be applied to the training dataset during model development.

Model performance will be assessed using the following classification metrics:

- a. Accuracy
- b. Precision
- c. Recall (Sensitivity)
- d. F1-score
- e. Area Under the Receiver Operating Characteristic Curve (AUC-ROC)

Comparative evaluation will be conducted to determine whether machine learning models provide improved predictive performance relative to the baseline logistic regression model.

Handling Class Imbalance

Since under-five mortality is expected to represent a minority class relative to survival cases, class imbalance may influence model training and predictive accuracy. To address this issue, class weighting techniques will be applied in logistic regression and machine learning algorithms where supported. Model evaluation will prioritize recall, F1-score, and AUC-ROC rather than relying solely on accuracy. If class imbalance significantly affects model performance, resampling techniques such as Synthetic Minority Oversampling Technique (SMOTE) will be applied to the training dataset only, in order to improve minority class detection while preventing data leakage.

Hyperparameter Tuning

To optimize predictive performance, hyperparameter tuning will be conducted using grid search combined with cross-validation. Parameters such as tree depth, number of estimators, and learning rate in ensemble models will be systematically adjusted to identify the optimal configuration. This process ensures that the final models are both accurate and generalizable to unseen data.

3.7 Ethical Considerations

This study utilizes secondary data obtained from the 2022 Tanzania Demographic and Health Survey and Malaria Indicator Survey. The dataset is publicly available upon approval and contains anonymized information with no personal identifiers. As such, the study does not involve direct interaction with human participants.

The data will be used strictly for academic purposes, and confidentiality will be maintained throughout the analysis process in accordance with DHS data usage policies.

3.8 Expected Outputs

The expected outputs of this study include:

- (a) A predictive model for under-five child mortality in Tanzania.
- (b) Identification of key demographic, maternal, household, and geographic predictors.
- (c) Comparative evaluation of statistical and machine learning models.
- (d) Policy-relevant recommendations for targeted child health interventions.

3.9 Project Timeline

The project will be conducted over a defined academic period and will include the following stages:

- (a) Proposal development and approval
- (b) Data acquisition and cleaning
- (c) Exploratory and descriptive analysis
- (d) Model development and evaluation
- (e) Report writing and submission
- (f) A detailed Gantt chart will be prepared to illustrate the timeline for each activity.

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